

Business Case Study: Blinkit Inventory & Sales Optimization

Project Title: End-to-End Quick Commerce Data Analysis

Organization: Blinkit (Quick Commerce Portfolio)

Tools Used: Python (Pandas, NumPy), Matplotlib, Seaborn, Jupyter Notebook

1. Executive Summary

Blinkit operates in the high-velocity "Quick Commerce" sector where inventory turnover and delivery speed are critical. The goal of this project was to analyze 8,500+ transaction records to identify revenue drivers, consumer health preferences, and outlet performance. Using Python's data science stack, I transformed raw operational data into strategic insights that could optimize inventory stocking and regional marketing spend.

2. Problem Statement

As a data-driven delivery platform, Blinkit needed to answer three critical business questions:

- **Consumer Behavior:** Is there a significant shift toward "Low Fat" vs. "Regular" products?
- **Regional Optimization:** Which outlet locations (Tier 1, 2, or 3) are generating the highest ROI?
- **Product Mix:** Which categories (Dairy, Snacks, Frozen Foods) are the core revenue drivers versus "filler" items?

3. Data Workflow (The Python Stack)

- **Data Wrangling (Pandas):** * Cleaned and pre-processed 8,523 rows of sales data.
 - Handled missing values in "Item Weight" and "Outlet Size" using statistical imputation.
 - Standardized categorical labels (e.g., merging "LF" and "low fat" into a single "Low Fat" category).
- **Exploratory Data Analysis (EDA):**
 - Utilized **NumPy** for complex mathematical aggregations of total sales and visibility scores.
 - Conducted correlation analysis to see if product visibility directly impacts sales volume.
- **Data Visualization (Seaborn/Matplotlib):**

- Created distribution plots to analyze price points across different product categories.
- Built heatmaps and bar charts to compare outlet performance across different years of establishment.

4. Key Performance Indicators (KPIs)

- **Total Revenue:** ~\$1.20M generated across all categories.
- **Average Sales per Item:** \$141.00.
- **Total Items Sold:** 8,523 individual SKU transactions analyzed.
- **Average Customer Rating:** 3.9/5.0.

5. Business Insights & Impact

- **Health Trends:** Discovered that "**Low Fat**" products contribute to **64.6% of total revenue**, indicating a strong consumer preference for healthy alternatives.
- **Location Strategy:** **Tier 3 cities** emerged as the top revenue stream (\$472K), proving that quick commerce has massive untapped potential outside of major metropolitan hubs.
- **Category Focus:** Identified "Fruits & Vegetables" and "Snack Foods" as the top-selling categories, contributing nearly **\$0.35M** combined.
- **Outlet Age:** Data showed that older outlets (established in 1985) maintained higher average sales, suggesting that customer loyalty and brand maturity are key long-term drivers.

Resume Bullet Points (Copy/Paste):

Blinkit Sales Analysis | Python (Pandas, Matplotlib, Seaborn)

- Engineered a Python-based data pipeline to process 8,500+ records, performing data cleaning and feature engineering to ensure 100% data integrity.
- Discovered that "Low Fat" products drive 64.6% of total revenue (\$1.2M), providing actionable insights for inventory stocking and health-focused marketing.
- Analyzed regional performance to identify Tier 3 cities as the highest revenue contributors (\$472K), aiding in geographic expansion strategy.